THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The equation $kx^2 + 4x + 2 = 0$ has real roots when: 1

(a) $k < 2$ only **(b)** $k \leq 2$ and $k \neq 0$

(c) $k > 0$ **(d)** $k \geq 2$

2. If α, β are roots of $x^2 - 7x + 12 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ 1

(a) 84 **(b)** $\frac{12}{7}$ **(c)** $\frac{7}{12}$ **(d)** $-\frac{7}{12}$

3. If the roots of $x^2 - 10x + 21 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 100 **(b)** 58 **(c)** 79 **(d)** 142

4. If the roots of $x^2 - 14x + k = 0$ differ by 6, then $k =$ 1

(a) 43 **(b)** 40 **(c)** 58 **(d)** 80

5. For which value of k does $x^2 + kx + 25 = 0$ have equal roots (take $k > 0$)? 1

(a) 5 **(b)** 10 **(c)** 25 **(d)** 20

6. If $x = -1$ is a root of $kx^2 + 4x + 6 = 0$, then $k =$ 1

(a) -2 **(b)** 2 **(c)** -6 **(d)** 10

7. The equation $kx^2 + 8x + 4 = 0$ has real roots when: 1

(a) $k > 0$ **(b)** $k \leq 4$ and $k \neq 0$

(c) $k < 4$ only **(d)** $k \geq 4$

8. If α, β are roots of $x^2 - 9x + 20 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ 1

(a) $\frac{20}{9}$ **(b)** 180 **(c)** $\frac{9}{20}$ **(d)** $-\frac{9}{20}$

9. **Assertion (A):** The equation $x^2 - 4x + 5 = 0$ has real and distinct roots. 1

Reason (R): If $b^2 - 4ac > 0$, the roots of $ax^2 + bx + c = 0$ are real and distinct.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The roots of $x^2 - 5x + 6 = 0$ are 2 and 3. 1
Reason (R): A quadratic equation has at most two roots.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If α, β are the roots of $x^2 - 8x + 7 = 0$, find $\alpha^2 + \beta^2$. 2

12. The sum of a number and its reciprocal is $\frac{10}{3}$. Find the number. 2

13. Find the values of k for which $kx^2 + 8x + 8 = 0$ has real and equal roots. 2

14. If α, β are the roots of $x^2 - 9x + 8 = 0$, find $\alpha^2 + \beta^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{x+1}{x-1} + \frac{x-1}{x+1} = \frac{5}{2}$. 3

16. For what values of k does $x^2 + 9x + k = 0$ have real roots? Find the greatest integer value of k . 3

17. The sum of the squares of two consecutive odd positive integers is 202. Find them. 3

OR

Solve for x: $\frac{x+3}{x-3} + \frac{x-3}{x+3} = \frac{17}{4}$. 3

18. For what values of k does $x^2 + 11x + k = 0$ have real roots? Find the greatest integer value of k. 3

OR

The sum of the squares of two consecutive odd positive integers is 290. Find them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A train travels 360 km at a uniform speed of x km/h. Had the speed been 5 km/h more, it would have taken 1 hour less for the same journey. 5
(i) Write the equation for the given condition.
(ii) Reduce it to $x^2 + 5x - 1800 = 0$ and find its discriminant.
(iii) Find the original speed of the train.
(iv) Find the time taken at the original speed and at the increased speed, and verify the 1-hour difference.

20. Two taps running together fill a tank in $\frac{18}{5}$ hours. The larger tap alone takes 3 hours less than the smaller one. Find the time each takes alone. 5

OR

A train travels 1080 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 3 hours less for the journey. Find the original speed. 5

END OF QUESTION PAPER — Total: 40 Marks